

Is There Depth Left to Fold?

Training-Free Structural Compression of a 1-Bit 27B Model, Measured End-to-End on a Laptop

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Abstract

Extreme weight quantization has produced open models that carry 27B-class capability at 1.125 bits per weight. We ask whether such models still contain structural slack, and whether it can be harvested with no training, no calibration-based weight updates, and byte-identical surviving weights, entirely on one 64 GB consumer laptop. Working on a public 1-bit 27B hybrid-attention model, we (1) map depth redundancy at four granularities and measure an additive damage-composition law; (2) ship a byte-identical artifact that removes 15.4% of the language model’s bytes while keeping GSM8K at reference parity, at a cost of 4.5 macro points on a 500-item four-task benchmark; (3) show the format stores a fully derivable bias plane, worth a measured -420 MB of disk and resident memory via kernels that re-derive it in-register with unchanged logits; (4) demonstrate the first training-free, in-format repair of a folded 1-bit model: a single closed-form ridge solve whose requantization flips 1% of sign bits and recovers $+1.0$ macro at identical bytes, together with the boundaries of the regime where it holds; and (5) document six independent cases in which proxy screens (full-vocabulary teacher-forced KL, two evaluation regimes) mispredicted benchmark outcomes, including an evolutionary search that wins the screen frontier and loses the benchmark. The pattern yields a per-operator-family law: deleting redundant structure has a shallow screen-to-benchmark damage slope, while densely perturbing every token’s computation has a steep one. Replication on a dense 8B sibling shows the structural laws are architecture-general while the harvestable slack is a property of over-provisioned depth. In this regime, screens propose and only benchmarks dispose.

1 Introduction

On-device inference economics are set by memory: at small batch sizes, generation speed is bounded by the bytes streamed per token. Weight formats have accordingly been pushed to one and two bits per weight, and recent open releases show a 27B model retaining roughly 90% of its full-precision benchmark average at 1.125 bits per weight. Such models are usually treated as terminal artifacts: the quantizer has spent the redundancy budget, and further compression is assumed to require retraining.

This paper asks what remains after extreme quantization, under deliberately austere rules: no training, no calibration-based weight updates, and, for the main result line, surviving weights byte-identical to the original pack. These constraints make every measured damage delta attributable purely to the structural edit; they make claims verifiable by byte comparison, logit equality, and per-item benchmark reproduction; and they respect a practical reality of the 1-bit regime, namely that the weights were produced by a proprietary transformation that nobody downstream can re-run. All experiments ran on one 64 GB Apple Silicon laptop.

Table 1 summarizes the contributions; the sections that follow give one result each.

2 Setup

Model. A publicly released 1-bit (binary, group-128 affine) 27B hybrid-attention causal LM: 64 blocks, 75% linear attention, 5120 hidden, 248K vocabulary, 4.71 GB of language-model weights. Weights take values $\pm s_g/2$ with one f16 scale s_g per 128 weights. A dense sibling (8B, 36 blocks, all full attention, same format) serves for generality tests.

contribution	headline number
Redundancy map, 4 granularities	additive composition, cross-pool tax 1.09–1.13×; replicated on a second architecture
Byte-identical artifact	−709 MB (−15.4%), macro .7963 vs. .8413, GSM8K unchanged at .910
Derivable bias plane	$b = \text{f16}(-s/2)$ verified on every group; −420,225,024 B resident memory, logits unchanged
In-format repair	one ridge solve, 1% sign flips, +1.0 macro at identical bytes; scope mapped
Screens vs. benchmarks	six documented divergences; per-operator-family damage-slope law

Table 1: Contributions.

Measurement ladder. Table 2. The benchmark is the only decision gate. Calibration data is disjoint from all probe sets; nothing is fitted on probes.

level	metric	size	role
screen	full-vocab teacher-forced KL, 2 regimes	24 probes/regime	shortlist only
confirmation	same, both regimes	100 probes/regime	survivor check
benchmark	GSM8K, MATH-500, IFEval, MMLU-Redux	500 items, seeded	decision gate

Table 2: The measurement ladder. On-policy probes are the model’s own reasoning traces; off-policy probes are held-out prose. Both regimes are mandatory (§8).

Operators. Structural operators delete whole blocks, individual attention or MLP sublayers, or whole GQA KV-groups; every surviving tensor is byte-identical. Two authorized value-modifying arms (§6, §7) are reported separately throughout.

3 The Redundancy Map and the Damage Law

Screening all 64 blocks singly gives a consistent shape: first and last blocks are catastrophically load-bearing; slack concentrates at mid-depth; and the two regimes disagree about which mid-depth blocks are cheapest (block 1 looks droppable on-policy and is ruinous off-policy, divergence #1 of §8). The shape recurs at sublayer granularity (deep linear-attention sublayers are individually cheap; the attention and MLP halves of a block can be removed independently) and at head granularity (early full-attention blocks surrender KV-groups almost freely at the logit level, a GQA over-provisioning invisible to coarser operators).

Composition is near-additive: across operator types and both models, the KL of a deletion set is 0.97–1.20× the sum of its singles, with the tax growing for same-type stacking. Table 3 gives the benched byte-identical ladder. The composed artifact dominates the naive six-block point, and task damage is non-uniform: GSM8K is unchanged while MMLU absorbs −8; folding removes diffuse breadth, not the reasoning spine.

config	−MB	GSM8K	MATH500	IFEval	MMLU-R	macro
reference (unfolded)	0	.910	.720	.930	.805	.8413
$k2$: 2 blocks	118	.920	.660	.880	.790	.8125
$k4$: 4 blocks	236	.910	.680	.920	.760	.8175
$k6$: 6 blocks	360	.880	.640	.860	.725	.7762
$k8$: 8 blocks	478	.870	.720	.790	.620	.7500
folded-709 ($k4 + 3$ attn subs)	709	.910	.670	.880	.725	.7963
search champion (§8)	365	.880	.610	.830	.730	.7625
head promotion (§8)	739	.850	.630	.830	.735	.7612

Table 3: Byte-identical arm, 500-item bench; sizes vs. the 4.71 GB pack. Below the rule: two probes that screened well and failed the bench. The folded-709 pack reproduces its benched configuration per-item (500/500).

4 Measured Negatives

Byte-identity permits several further axes; all were tested and rejected (Table 4).

axis	test	outcome
width	channel-magnitude profiles, all layers	flat; no low-salience tail at 1 bit
vocabulary (input)	held-out OOV mass	fat tail; rows load-bearing
vocabulary (out-put)	emission-mass keep-sets	teacher-forced logits unchanged, generation broken: +26% length, $2.3\times$ truncations (divergence #2); mixed-domain sets need 87–98% of rows
merging	sign-election (averaging analog) and format-native 2-bit promotion	both lose to plain dropping at matched bytes
sign plane	compression entropy	8.000 bits/byte; incompressible

Table 4: Rejected axes, byte-identical arm.

5 A Free Format Win: the Derivable Bias Plane

The pack stores, per group, both a scale and a bias, but the 1-bit mapping fixes $b = \text{f16}(-s/2)$; we verified the identity on every group in the model (the vendor’s report notes the redundancy and defers the fix). Deleting the plane costs nothing on disk (−420 MB, logits unchanged). We then implemented the deferred fix: a new kernel mode derives the bias in-register. All matmul routes and dequantization produce logits equal to the stored-bias path, on synthetic parity batteries and on the full 27B, and resident memory drops by the plane’s full 420,225,024 bytes. The derivation trades one multiply for a removed memory stream in bandwidth-bound kernels, so it is never slower.

6 Flagged Arm I: Quantizing the Metadata

With the bias plane derived, the model is a 3.77 GB sign plane plus 420 MB of f16 scales holding all continuous information. Quantizing scales is value-modifying, hence flagged and reported apart (Table 5). At 8 bits it is free within noise and composes cleanly with folding; the combined artifact reaches 3.82 GB (−18.9%) at .7900. The precision knee at 4 bits is benchmark-visible only: the screen sits $76\times$ below the gentlest block drop while the bench shows $k2$ -class damage (divergence #4).

config	−MB	macro	verdict
8-bit scales (unfolded)	612	.8363	free within noise (−0.5)
combined: folded-709 + 8-bit scales	888	.7900	3.82 GB; composes cleanly
4-bit scales (unfolded)	717	.8125	knee: −2.9, $k2$ -class damage
folded-709 + repair (§7)	709	.8063	+1.0 at identical bytes
combined + repair	888	.7825	worse: no composition (div. #5)
$k6$ + repair	360	.7750	flat: no deep-fold reach

Table 5: Value-modifying arms, 500-item bench. All rows keep the sign planes of non-designated tensors untouched; sizes vs. the 4.71 GB pack.

7 Flagged Arm II: Training-Free In-Format Repair

Folding damage is the loss of dropped modules’ contributions to the residual stream. The 1-bit format offers an unusual repair channel: a module’s effective weights are fully determined by its sign bits and group scales,

so a weight-space correction can be requantized into the format at zero byte cost. We fit each dropped cluster’s *linear shadow*, the best static linear map ΔW from a surviving downstream projection’s input to the stream deficit, with a single ridge solve over 64 calibration sequences; we cap $\|\Delta W\|$ at $0.25\|W\|$ so requantization noise stays second-order; and we requantize $W + \Delta W$ per group (new scale $2\lceil w^* \rceil$, new signs $\text{sign}(w^*)$).

On folded-709 this flips 0.8–1.7% of sign bits in three projections, drifts scales under 4.5%, and recovers +1.0 macro at identical bytes (Table 5), with a broad task pattern and no generation pathologies. To our knowledge this is the first training-free, in-format repair of a 1-bit model: the sign/scale lattice around a folded model contains strictly better points, reachable by one closed-form solve. Its scope is narrow and mapped in the same table: a diagonal-gains variant fails (downstream RMSNorms already absorb static scale effects, which is why folding is cheap in the first place); stacking repair on the scale-quantized artifact reverses its sign despite a screen profile indistinguishable from the confirmed run; and on a deeper fold the correction cap saturates and the effect vanishes.

8 Screens Propose, Benchmarks Dispose

Six times under one fixed protocol, proxy metrics mispredicted the benchmark (Table 6).

#	configuration	screen verdict	bench verdict
1	single mid-depth block	cheap (on-policy only)	ruinous off-policy
2	vocabulary keep-set	logits unchanged	+26% length, 2.3× truncations
3	evolutionary search champion	dominates frontier, both regimes	−1.4 below byte-matched <i>k6</i>
4	4-bit scales	76× below <i>k2</i>	<i>k2</i> -class damage (−2.9)
5	combined + repair	matches a confirmed repair profile	sign reversed (−0.75)
6	8 KV-groups on folded-709	sub-additive, below <i>k6</i> twice	−3.5 for 30 MB, below <i>k6</i>

Table 6: Six screen-vs-benchmark divergences. #3: a $(1+\lambda)$ search over 311 per-block operator genomes whose KL dominance grows to 40% off-policy at the aggressive tier, and whose champion still loses the bench.

Instances 3–6 support a compact law: operator families have characteristic screen-to-benchmark damage slopes. Deleting redundant structure is shallow-sloped; the removed computation was substitutable, and KL rank-orders such configurations correctly. Densely perturbing every token’s computation (metadata noise, KV-group removal, requantization residue) is steep-sloped: teacher-forced KL scores one forced step at a time and systematically underweights how dense small perturbations compound across a long generation. Screen numbers are therefore comparable only within an operator family, and no screen, however well validated, substitutes for the benchmark.

9 Generality

Table 7 tests every law on the dense 8B sibling. The map’s shape replicates; the harvestable magnitude does not. A cross-model KL→damage curve was tested by interpolation and rejected. Foldability is a property of over-provisioned depth, not of 1-bit models per se: the 27B folds because it is deep for its capability, while the format redundancy ($b = -s/2$, verified on all 254 tensors) is family-wide.

10 Limitations

One model family and format carries the headline results. $N=500$ per bench bounds resolution to roughly ± 1.8 macro points at 1σ , so sub-point deltas are directional on single-bench evidence. The 16K thinking budget confounds the damage floor. The six divergences are demonstrations under one protocol, not a survey. The repair result is an existence proof with mapped boundaries, not a general method. Everything was produced and verified on one 64 GB laptop, which is both the point and the ceiling: search breadth, sampling repeats, and bench sizes were budgeted accordingly.

law	27B (hybrid)	8B (dense)
boundary blocks catastrophic	yes	yes (block 0: 7.7 nats)
mid-depth slack	yes	yes
two-regime disagreement	yes	yes
additive composition tax	1.09–1.13×	0.97–1.20×
cheapest single (on-policy KL)	.0033	.0198 (6×
4-cheapest-blocks bench cost	−2.4	−14.8
cross-model damage curve	rejected (predicted −7 to −9, measured −4.3)	

Table 7: Generality: structural laws replicate, harvestable slack does not.

11 Conclusion

A model that survived 14× weight compression still contained a fifth of its bytes in removable or derivable form at ≤ 4.5 points of macro cost with reasoning intact, a free 420 MB format plane, nearly-free metadata precision down to 8 bits and not one bit further, and a repairable neighborhood in its own sign/scale lattice. Extracting all of it required no training: only deletion, derivation, one linear solve, and the discipline of never believing a screen. Code and artifacts will be released upon publication.